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TRƯỜNG ĐẠI HỌC CÔNG NGHỆ THÔNG TIN
KHOA KHOA HỌC MÁY TÍNH



SELF-STUDY

Topic: RNN

GVHD: My self
SV thực hiện: Lại Khánh Hoàng

Tp. Hồ Chí Minh

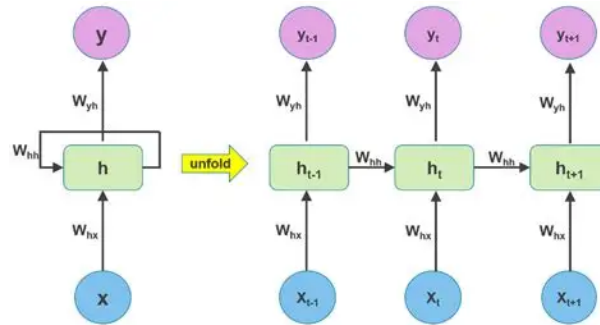


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1 Equation

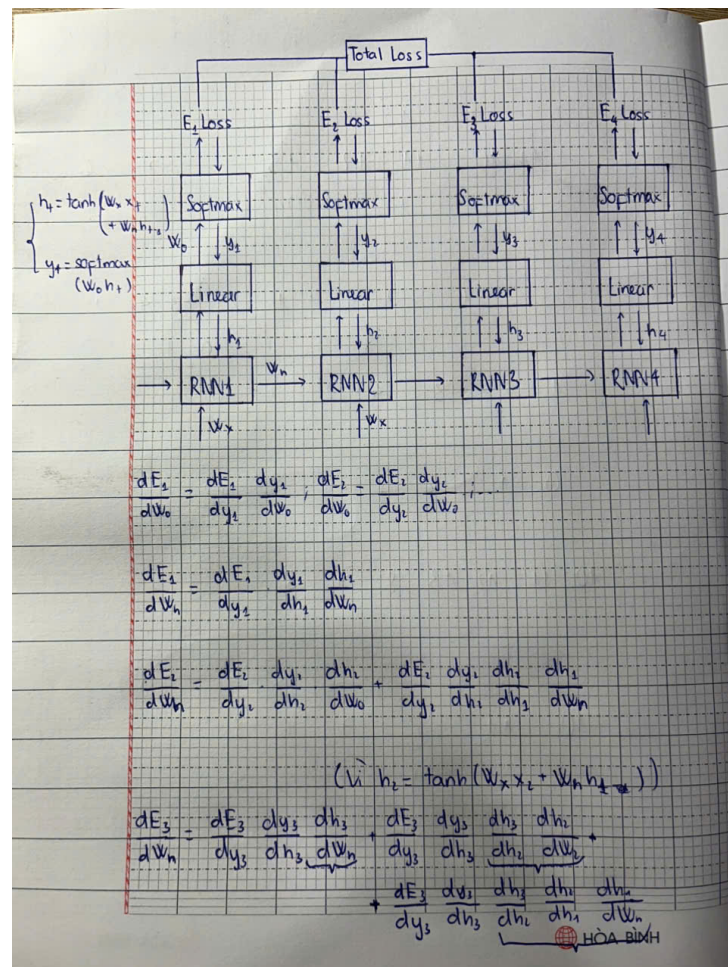


$$\begin{cases} y_t = W_{yh}h_t \\ h_t = f(h_{t-1}, x_t) \left(\begin{array}{l} \text{in case } f=\tanh \\ = \tanh(W_{hh}h_{t-1} + W_{hx}x_t) \end{array} \right) \end{cases}$$

2 Backpropagation Through Time (BPTT)

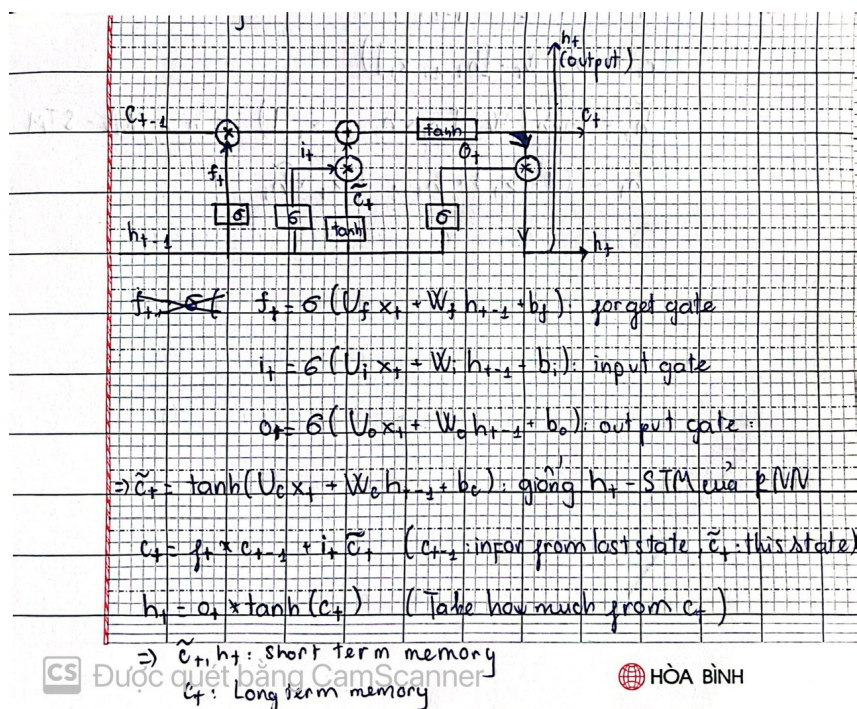
For all time step:

$$\begin{cases} \frac{dE}{dW_o} = \sum_{t=1}^T \frac{dE_t}{dW_o} \\ \frac{dE}{dW_h} = \sum_{t=1}^T \frac{dE_t}{dW_h} \\ \frac{dE}{dW_x} = \sum_{t=1}^T \frac{dE_t}{dW_x} \end{cases}$$

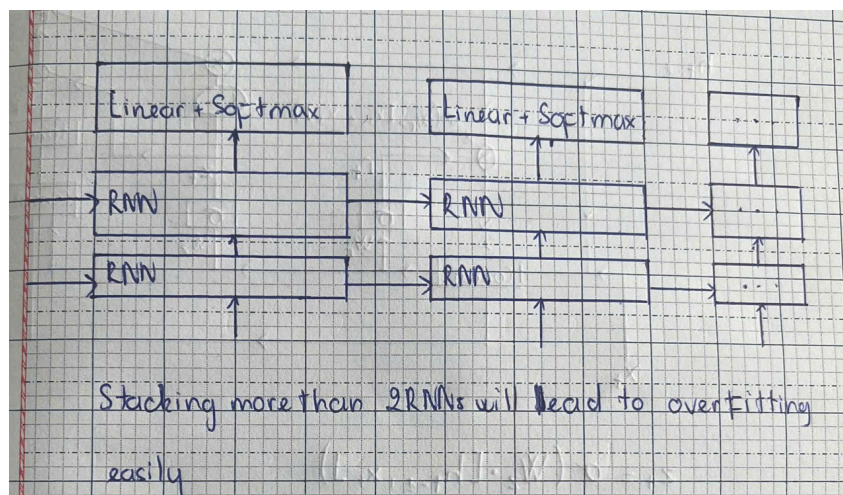




3 LSTM



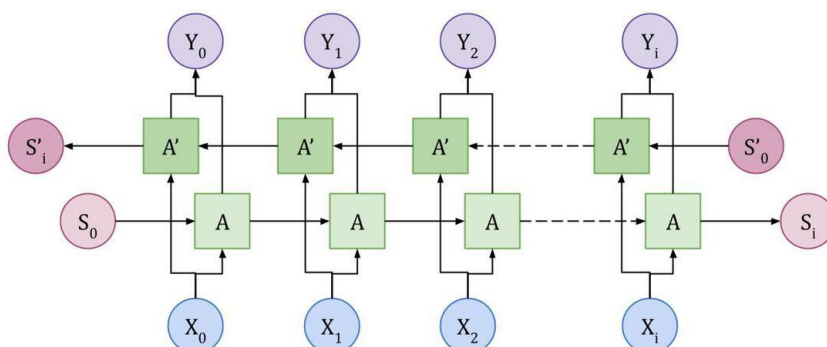
4 Stacked-RNNs



5 Bi-Directional LSTMs

Rather than encoding the sequence in the forward direction only, we encode it backward direction as well, and concatenate the results from both the forward and backward LSTM at each time step. The encoded representation of each word now understands the word *before* and after *that* specific word.

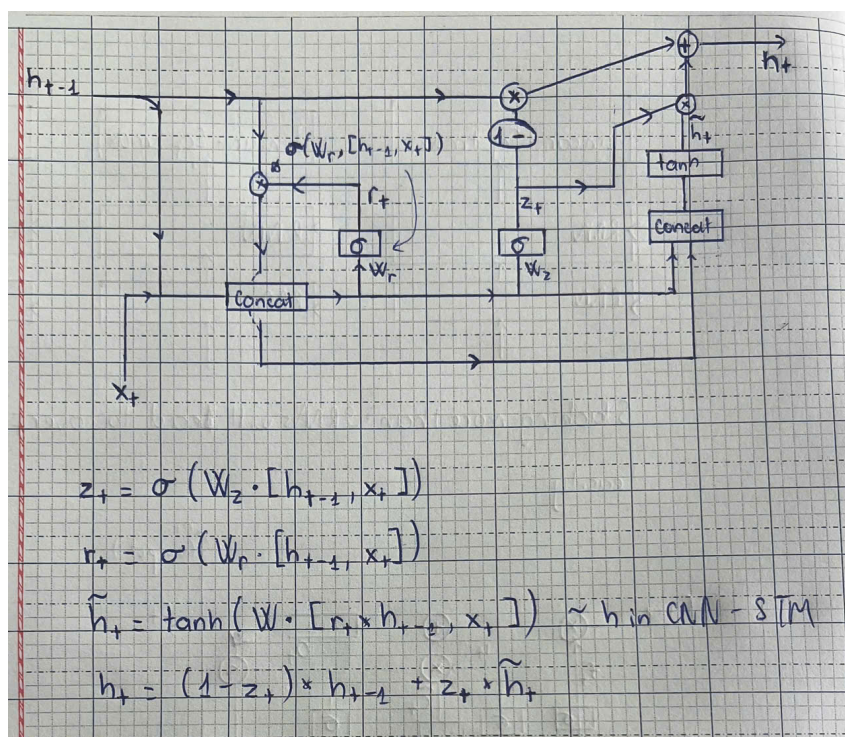
LSTMs can only be used for encoding a sequence, not for generating words.



6 GRUs (Gated Recurrent Units)

Advantage of GRU:

- GRUs is less computationally expensive than LSTMs, and they still perform on par with LSTMs, so they train faster
- For larger dataset and longer sequence, LSTMs are able to capture better memory. But the difference is only little
- Usually used for short sequences



7 LSTM-CNN

- Is an LSTM architecture specifically designed for sequence prediction problems with spatial inputs, like images or videos
- The CNN-LSSM architecture involves using CNN for feature extraction on input data combined with LSTMs to support sequence prediction.

Example:

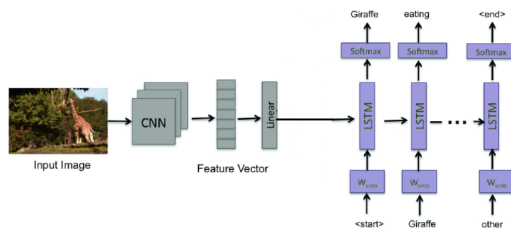
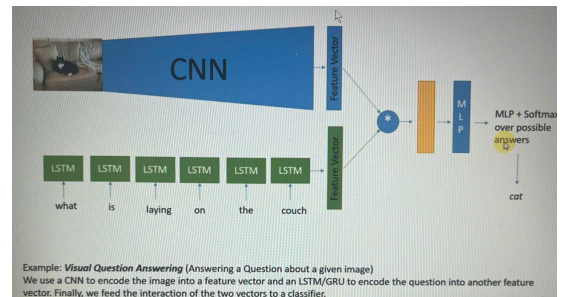


Figure 2: Baseline encoder-decoder architecture using a pretrained ResNet-50 convolutional neural network encoder and long short term memory model decoder

Hình 1: Image Captioning



Hình 2: QA